MULTI-SITE SOLAR PV PORTFOLIO (32.0 kW)

Date: 22-Sep-2026

Portfolio: Commercial Multi-Branch Portfolio (2 Sites, 2 Buildings) | Location: Karnataka | Grid-Tied LT Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
32.00 kW	**50,737 kWh**	**Rs. 1,395,250**	**4Y 5M**
32.45 kWp Installed	CUF 18.1% | PR 75%	Rs. 13.95 Lakhs	Profit: Rs. 150.11L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	Commercial Multi-Branch Portfolio
Contact Mobile / Phone	+91 99999 00000
Location / Address	Karnataka
Contact Email Address	Not Specified
Design Path Selected	Rooftop Area Analysis
Approved Sanctioned Load (CD)	25 kW (KERC / ESCOM Tariff Order)
System Target Capacity	32.00 kW (2 Sites Portfolio)
Total Installed Array Rating	32.45 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	59 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	3775.00 Sq.Feet (350.71 sq.m)
Utilized Usable Rooftop Area	3433 Sq.Feet (318.94 sq.m)
First Year Annual Generation	50,737.28 kWh
25-Year Cumulative Generation	1,127,274.00 kWh
Estimated Payback Period	4 Years & 5 Months
Post-Payback Profitability (25 Yrs)	Rs. 15,010,579.50 (INR 150.11 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 6,735,847.18 (INR 67.36 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 924 Tonnes CO2 | Tree Equivalence: 1478 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Statutory ceiling enforced under KERC SRTPV Regulations. System automatically sized to maximum allowable 20.0 kW.

[PORTFOLIO] Multi-Site Portfolio Allocation & Regulatory Schedule

Aggregated LT solar portfolio overview, site-wise capacities, and statutory compliance

TOTAL SITES	TOTAL CAPACITY	TURNKEY CAPITAL	ANNUAL YIELD
PORTFOLIO	AGGREGATE	12-ITEM BOM	CLEAN POWER
2 Sites	**32.00 kW**	**Rs. 1,395,250**	**50,737 kWh**
2 Buildings Total	59 Panels (550W)	Rs. 13.95 Lakhs	1st Year Aggregate

[ALLOCATION] Site-by-Site Allocation & Engineering Capacity Matrix

Independent LT net-meter installations conforming to 20 kW per-connection statutory ceiling

Sl.	Site Name & Location	Bldgs	Gross Area	Capacity	Inverter	1st Yr Yield	Turnkey Cost
01	Shivamogga Branch (Shivamo..	1 Nos.	2,200 sq.ft	20.00 kW	14 kW 3-Ph	31,711 kWh	Rs. 877,300
02	Mangaluru Branch (Mangalur..	1 Nos.	1,575 sq.ft	12.00 kW	9 kW 3-Ph	19,026 kWh	Rs. 517,950
	TOTAL PORTFOLIO SUMMARY		**3,775 sq.ft**	**32.00 kW**	**23 kW Total**	**50,737 kWh**	**Rs. 1,395,250**

KERC & ESCOM LOW TENSION (LT) 20 kW REGULATORY COMPLIANCE MANDATE

Under Karnataka Electricity Regulatory Commission (KERC) Rooftop Solar PV regulations and ESCOM interconnect guidelines, individual Low Tension (LT) net-metered service connections are subject to a statutory capacity ceiling of 20.0 kW per consumer installation / meter account. Interconnections above 20 kW necessitate High Tension (HT 11 kV) infrastructure, dedicated transformer bays, and elevated statutory approvals.

To maintain full regulatory compliance while meeting multi-site commercial / institutional power demand, this portfolio is architected into discrete, independent LT installations. Each individual site operates with its own dedicated bidirectional net-meter, solar inverter, ACDB protection board, and safety switchgear, while engineering specifications, equipment warranties, and turnkey financials are consolidated within this unified dossier.

LT PORTFOLIO STATUTORY & GRID INTERCONNECTION STANDARDS

Metering Architecture: Independent Bi-Directional TOD Net-Meters per site (1 - 20 kW LT Class)

Anti-Islanding Protection: Mandatory automatic grid disconnection upon utility outage (IEC 62116 / IEEE 1547 compliant)

Switchgear & Isolation: Dedicated IP65 ACDB with Type-II AC/DC SPDs & 10 kA Breaking MCBs per connection

DISCOM Permitting: Separate online DISCOM portal filings with aggregated turnkey EPC execution and common monitoring

[ARRAY] Shivamogga Branch -- Building 1: Layout & Array Sizing

Site: Shivamogga Branch (Shivamogga, Karnataka) | Sub-Array: 20.00 kW | Site Inverter: 14 kW

Building / Array Identification	Shivamogga Branch -- Main Wing (Rooftop Array)
Gross Rooftop Area	2200.00 Sq.Feet (204.39 sq.m)
Utilized Effective Rooftop Area	2014 Sq.Feet (187.11 sq.m)
Building Plant Target Capacity	20.00 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	37 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	14 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (14 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	14 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Deye, Delta Solar
Derated AC Design Power (70% Factor)	14.00 kW AC (from 20.00 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	63A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	19.48 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[ARRAY] Mangaluru Branch -- Building 1: Layout & Array Sizing

Site: Mangaluru Branch (Mangaluru, Karnataka) | Sub-Array: 12.00 kW | Site Inverter: 9 kW

Building / Array Identification	Mangaluru Branch -- Office Block (Rooftop Array)
Gross Rooftop Area	1575.00 Sq.Feet (146.32 sq.m)
Utilized Effective Rooftop Area	1419 Sq.Feet (131.83 sq.m)
Building Plant Target Capacity	12.00 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	22 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	9 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (9 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	9 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Greentech Energy
Derated AC Design Power (70% Factor)	8.40 kW AC (from 12.00 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	40A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	11.69 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[SITE BOM] Site 1: Shivamogga Branch -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Shivamogga Branch (20 kW)

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Mono-PERC (Mono-facial)	37 Nos.	Rs. 370,000
02	On-grid inverter	14 kW, three-phase (Deye or equiv.)	1 No.	Rs. 147,000
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 95,000
04	DC solar cable	6 sq.mm copper, PV-rated	220 m	Rs. 18,700
05	AC cable	35 sq.mm copper	110 m	Rs. 49,500
06	MC4 connectors	PV-rated connector pairs	37 pairs	Rs. 11,100
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 18,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 18,000
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 9,000
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 8,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 38,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 95,000
		TOTAL PLANT-SIDE CAPITAL COST (Shivamogga Branch)		**Rs. 877,300.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Shivamogga Branch)

Official DISCOM statutory filing fees and net-metering charges for this connection

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL ESTIMATED CAPITAL OUTLAY FOR SHIVAMOGGA BRANCH

Turnkey Plant BOM: Rs. 877,300 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 900,312 (Rs. 9.00 Lakhs)

[SITE BOM] Site 2: Mangaluru Branch -- Turnkey Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown for Mangaluru Branch (12 kW)

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Mono-PERC (Mono-facial)	22 Nos.	Rs. 220,000
02	On-grid inverter	9 kW, three-phase (Greentech Energy or equiv.)	1 No.	Rs. 97,500
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 55,000
04	DC solar cable	6 sq.mm copper, PV-rated	130 m	Rs. 11,050
05	AC cable	16 sq.mm copper	70 m	Rs. 16,800
06	MC4 connectors	PV-rated connector pairs	22 pairs	Rs. 6,600
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 12,500
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 12,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 5,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 18,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 55,000
		TOTAL PLANT-SIDE CAPITAL COST (Mangaluru Branch)		**Rs. 517,950.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget (Mangaluru Branch)

Official DISCOM statutory filing fees and net-metering charges for this connection

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL ESTIMATED CAPITAL OUTLAY FOR MANGALURU BRANCH

Turnkey Plant BOM: Rs. 517,950 | ESCOM Budget: Rs. 23,012 | Site Net Total: Rs. 540,962 (Rs. 5.41 Lakhs)

[PORTFOLIO TOTAL] Consolidated Portfolio Capital Cost & Grand Total Investment

Consolidated turnkey capital outlay across all portfolio sites and DISCOM budgets

TOTAL CAPACITY	TURNKEY BOM	ESCOM BUDGET	TOTAL CAPITAL
AGGREGATE	12-ITEM BOM	UTILITY	GRAND TOTAL
32.00 kW	**Rs. 13.95L**	**Rs. 0.46L**	**Rs. 14.41L**
2 Sites Sized	Rs. 1,395,250	Rs. 46,024	Rs. 1,441,274

[CAPEX MATRIX] Portfolio Site-by-Site Capital Investment Matrix

Comprehensive turnkey capital expenditure required for all sites

Sl.	Site Name & Location	Buildings	Capacity	Plant BOM Cost	ESCOM Budget	Total Outlay
01	Shivamogga Branch	1 Nos.	20.00 kW	Rs. 877,300	Rs. 23,012	Rs. 900,312
02	Mangaluru Branch	1 Nos.	12.00 kW	Rs. 517,950	Rs. 23,012	Rs. 540,962
GRAND TOTAL CAPITAL REQUIRED (ALL SITES)			**32.00 kW**	**Rs. 1,395,250**	**Rs. 46,024**	**Rs. 1,441,274**

** Marked installation(s) capped at statutory 20.0 kW Low Tension (LT) ceiling in accordance with KERC regulations.*

PORTFOLIO PROCUREMENT, EPC EXECUTION & DISCOM GRID CONNECTIVITY

All electrical installations conform to CEA Technical Standards for Distributed Generation and State Grid Code. Turnkey procurement bundles Tier-1 TOPCon modules and high-efficiency inverters under consolidated bulk pricing, while each site operates under its dedicated independent DISCOM net-meter account (RR number). This avoids High Tension (HT 11 kV) transformer infrastructure and maintains complete statutory compliance.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 1,395,250	**4Y 5M**	**Rs. 150.11L**	**Rs. 67.36L**
12-Item BOM Basis	Break-Even Year	Rs. 15,010,580	Alternative Asset

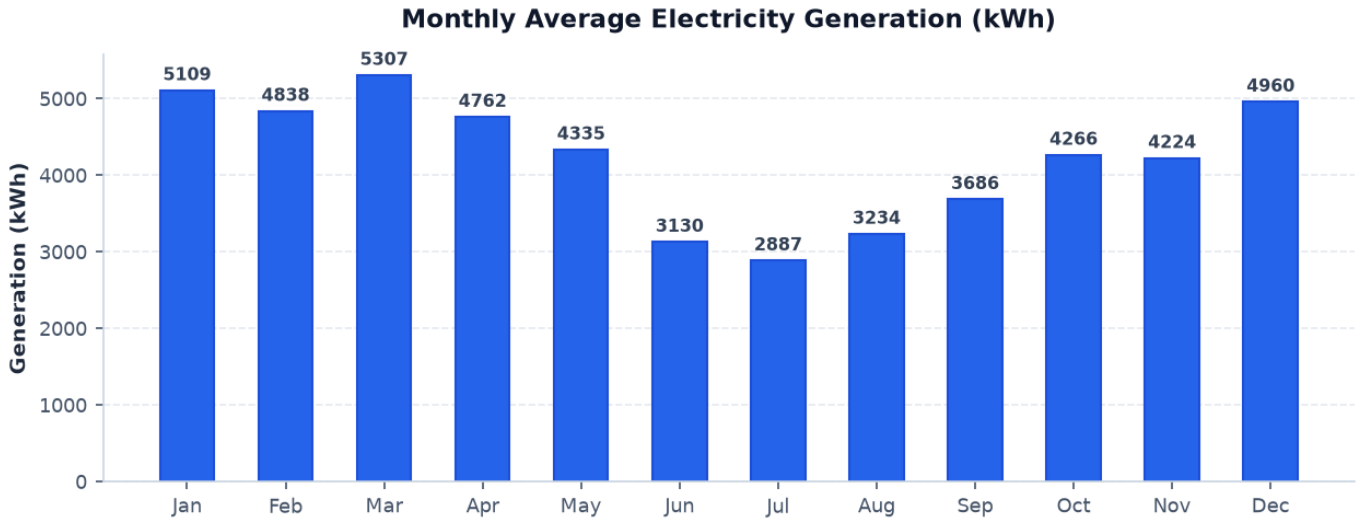
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	50,737	Rs. 8.05	408,435	19,200	389,235	389,235
02	50,230	Rs. 8.45	424,443	19,200	405,243	794,478
03	49,728	Rs. 8.87	441,084	19,200	421,884	1,216,362
04 *	**49,230**	**Rs. 9.32**	**458,827**	**19,200**	**439,627**	**1,655,988**
05	48,738	Rs. 9.78	476,658	19,200	457,458	2,113,446
06	48,251	Rs. 10.27	495,534	19,200	476,334	2,589,780
07	47,768	Rs. 10.78	514,941	19,200	495,741	3,085,521
08	47,290	Rs. 11.32	535,328	19,200	516,128	3,601,649
09	46,818	Rs. 11.89	556,661	19,200	537,461	4,139,110
10	46,349	Rs. 12.48	578,440	19,200	559,240	4,698,350
11	45,886	Rs. 13.11	601,564	19,200	582,364	5,280,714
12	45,427	Rs. 13.76	625,076	19,200	605,876	5,886,590
13	44,973	Rs. 14.45	649,856	19,200	630,656	6,517,246
14	44,523	Rs. 15.18	675,860	19,200	656,660	7,173,906
15	44,078	Rs. 15.93	702,159	19,200	682,959	7,856,865
16	43,637	Rs. 16.74	730,484	19,200	711,284	8,568,149
17	43,201	Rs. 17.57	759,035	19,200	739,835	9,307,984
18	42,769	Rs. 18.45	789,081	19,200	769,881	10,077,866
19	42,341	Rs. 19.37	820,144	19,200	800,944	10,878,810
20	41,918	Rs. 20.34	852,603	19,200	833,403	11,712,213
21	41,498	Rs. 21.36	886,405	19,200	867,205	12,579,418
22	41,083	Rs. 22.43	921,500	19,200	902,300	13,481,719
23	40,673	Rs. 23.55	957,839	19,200	938,639	14,420,358
24	40,266	Rs. 24.73	995,774	19,200	976,574	15,396,932
25	39,863	Rs. 25.96	1,034,848	19,200	1,015,648	16,412,580

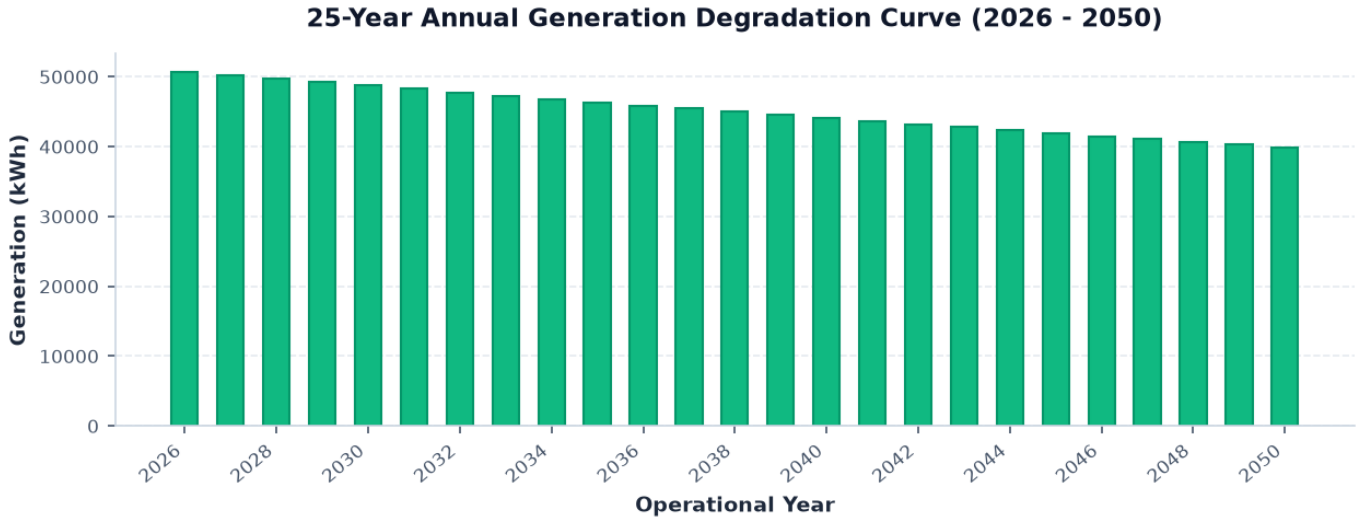
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

VoltRoof Designs
CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

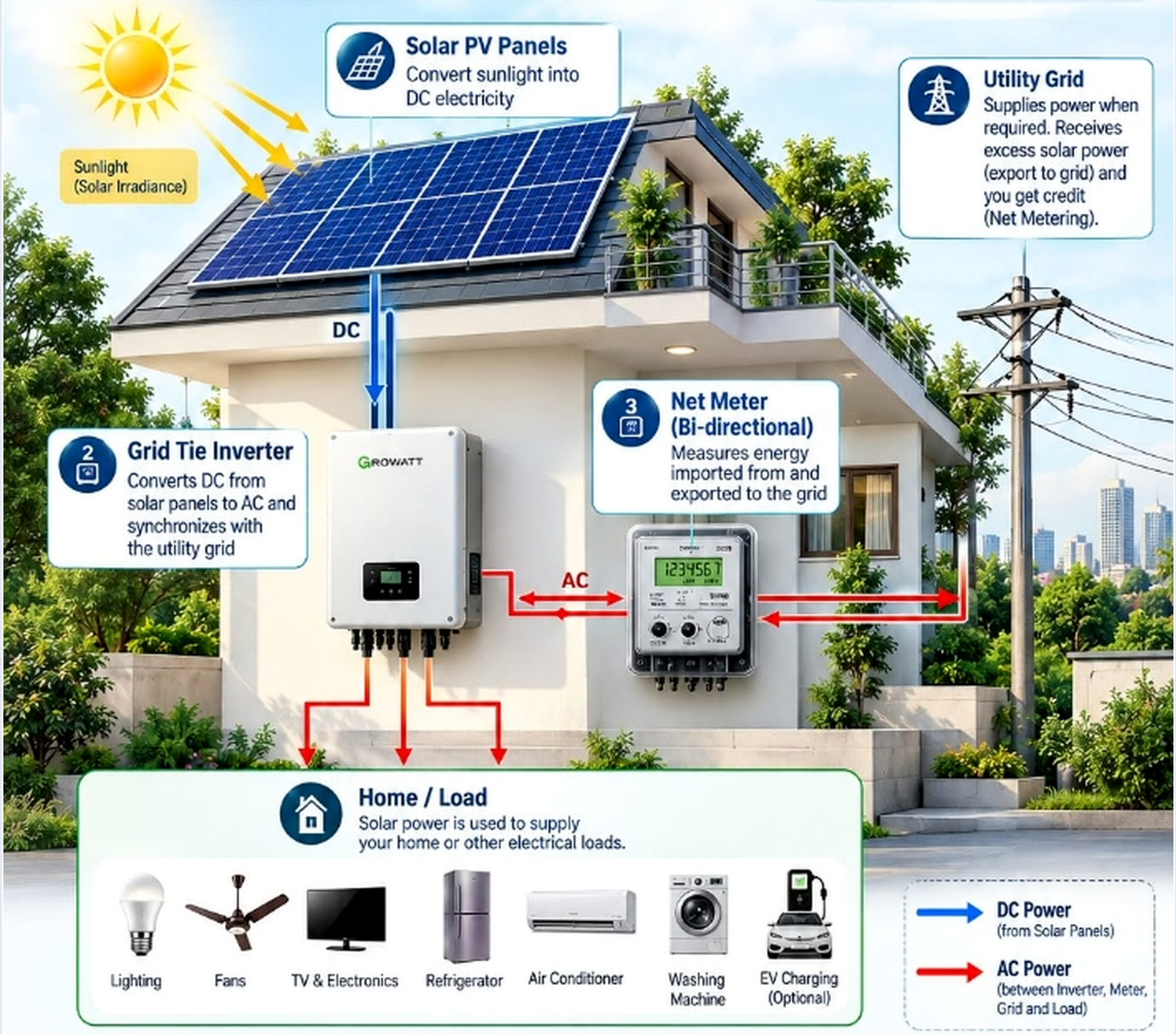
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow